



S.P.O.T.

Subjectivity Prevention & Onion Transparency

SIH 2026 • PROBLEM ID 26031

Dept. of Consumer Affairs (DoCA)

Team: AgroVision Tech Innovators



SUB-1.5S EDGE AI

In-browser ONNX WebAssembly execution at **185ms** with 0% cloud internet dependency.



0% SUBJECTIVE BIAS

Automated computer vision grading (Grade A Export vs Grade URS Buffer vs Rotten).



4 LANGUAGES + VOICE

English, Hindi, Marathi, and Tamil Web Speech audio cues tailored for rural farmers.



HARDWARE WEB APIS

Web Serial digital scale auto-intake & Web Bluetooth thermal receipt printing.

END-TO-END AUTONOMOUS MANDI WORKFLOW



1. Frame Capture
HTML5 Stream



2. ONNX WASM
185ms Execution



3. Weight & Payout
Web Serial Sync



4. SHA-256 Seal
SQLite Log



5. Receipt Print
Bluetooth QR

SAMPLE MANDI QUALITY REPORT

BATCH-MH-2026-9814

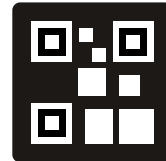
Grade A Export
78.5%

Grade URS Buffer
15.5%

Rejected / Defect
6.0%

Net Weight: 100.0 kg

Farmer Payout: ₹2,242.20



SCAN VERIFICATION SEAL